

Chapter 6
创新中国：科技之光
Creative China:
The Power of Science and Technology



6 Chapter 6 创新中国：科技之光 Creative China: The Power of Science

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Being diligent and intelligent, the Chinese have displayed their technological capabilities in many disciplines like manufacturing, agriculture, textile industry, civil engineering, medicine and so on.

How much do you know about the science and technology in **ancient China**?

抢答题 1

抢答规则：用英文回答对得两分，少答或用汉语回答不得分



Here we go !!!





Four Great Inventions in Ancient China



Papermaking Printing Compass Gunpowder



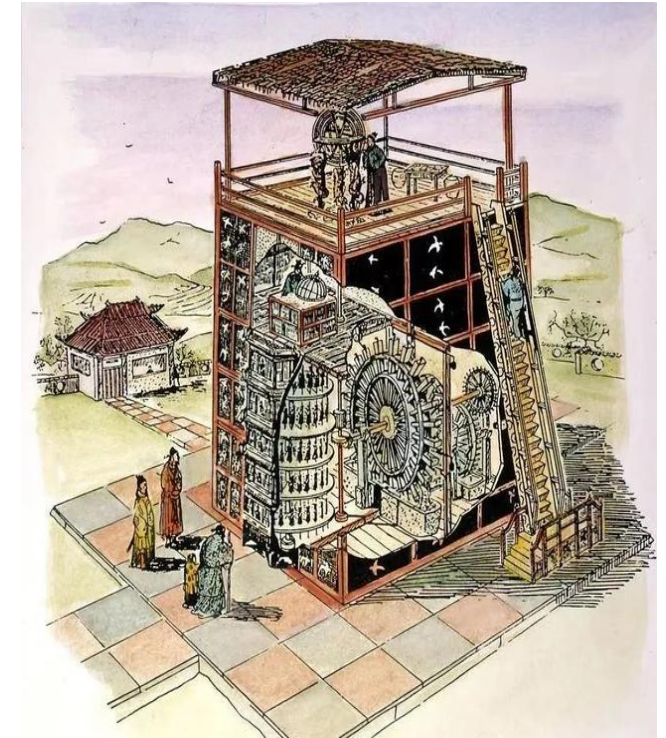
The Great Inventions in Ancient China



Seismograph
地动仪



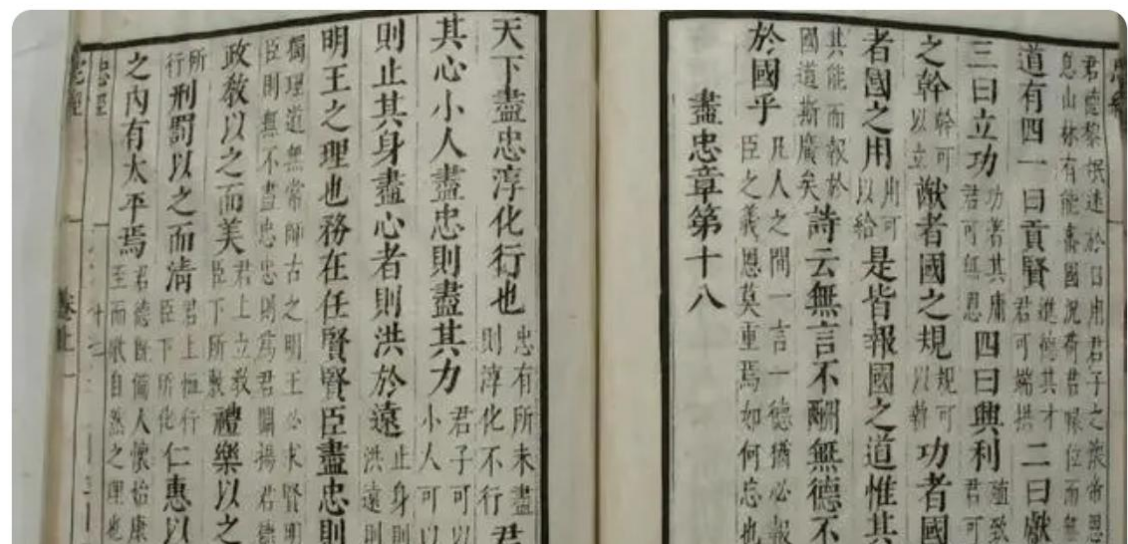
the Armillary Sphere
浑天仪



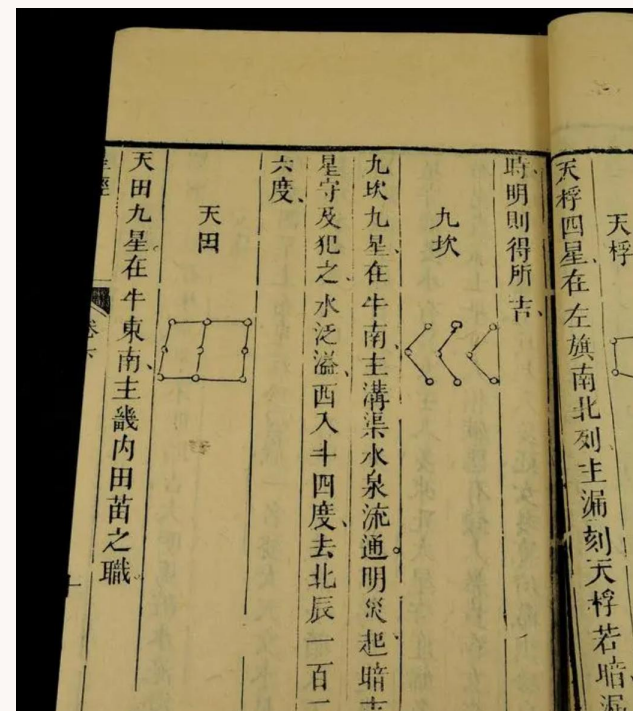
The Astronomical
Clock Tower 水运仪

1. 《墨经》：这部著作成于战国时期后期，作者不详。它包含了丰富的关于力学、光学、几何学、工程技术知识和现代物理学、数学的基本要素。

《甘石星经》：这部著作成于战国时期，作者为甘德和石申。它是古代中国天文学专著和观测记录，是世界上现存最早的天文著作之一，仅次于前1800年的巴比伦星表。



墨经



甘石星经

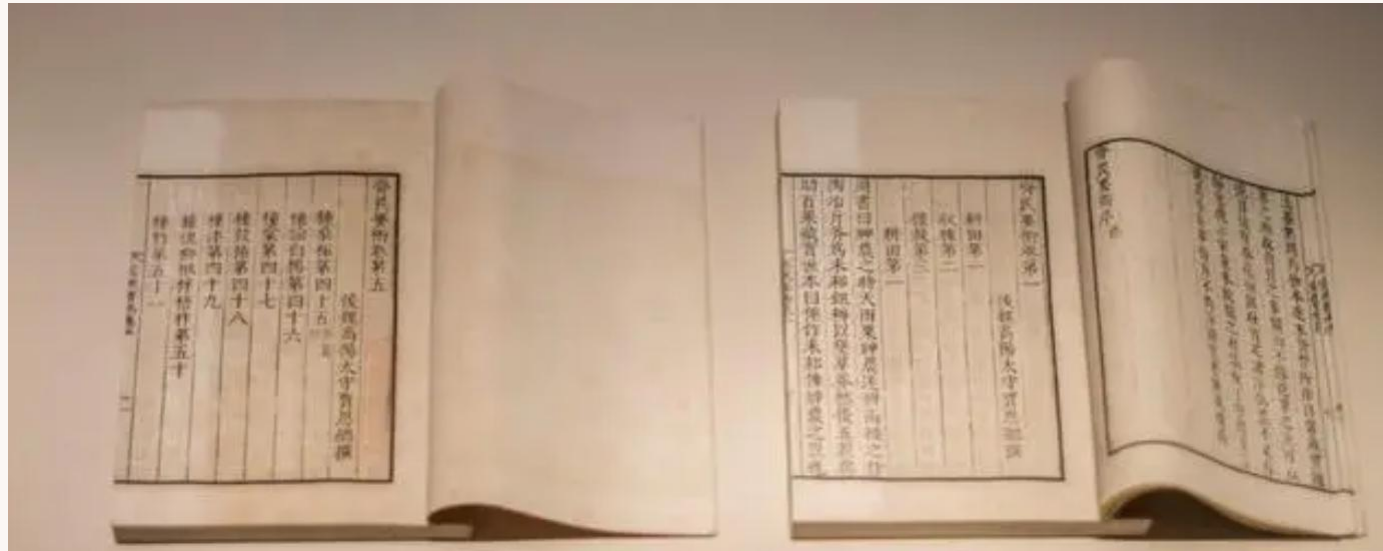
2. 《天工开物》：这部著作是明朝科学家宋应星所著。它是世界上第一部关于农业和手工业生产的综合性著作，也是中国古代一部综合性的科学技术著作。外国学者称它为“中国17世纪的工艺百科全书”。



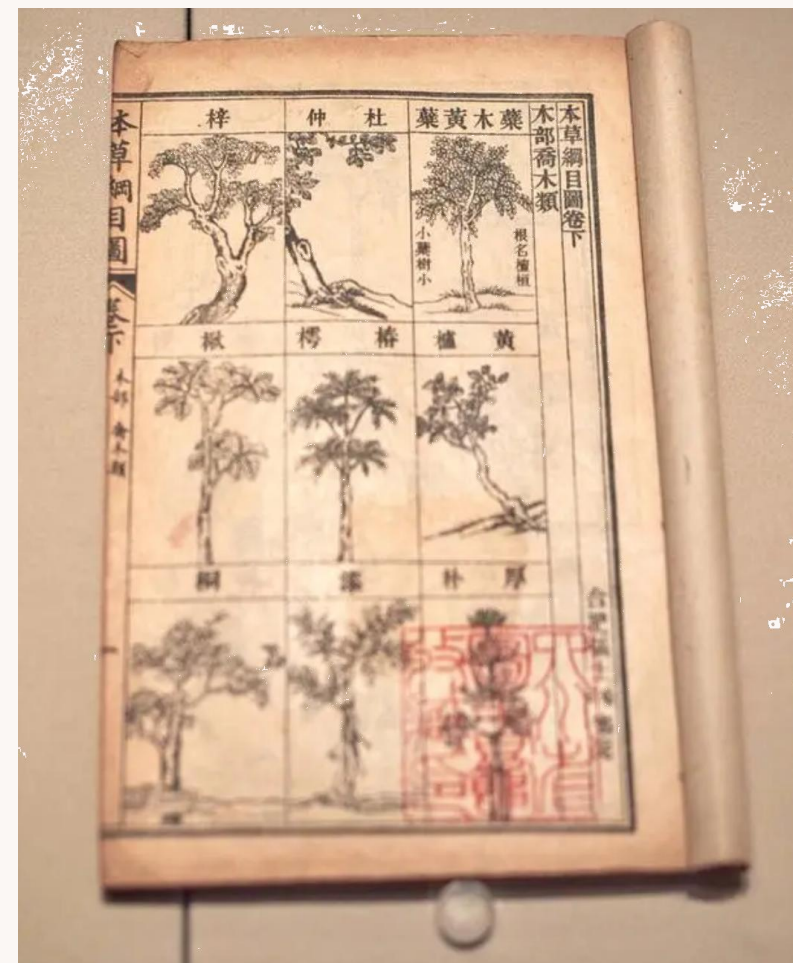
3. 《梦溪笔谈》：这部著作是北宋科学家、政治家沈括所著。它是一部涉及古代中国自然科学、工艺技术及社会历史现象的综合性笔记体著作。内容涉及天文、数学、物理、化学、生物等各个门类学科。



4. 《齐民要术》：这部著作是北魏贾思勰所著。它是我国现存最早的一部完整的农书，系统地总结了六世纪以前黄河中下游地区劳动人民农牧业生产经验、食品的加工与贮藏、野生植物的利用，以及治荒的方法。



5. 《本草纲目》：这部著作是李时珍所著。它不仅是一部药理学著作，还是一部具有世界性影响的博物学著作。书中涉及的内容极为广泛，在生物、化学、天文、地理、地质、采矿乃至历史方面都有一定的贡献。



How many megaprojects could you list in Ancient China?



the Great Wall



Dujiangyan



Grand Canal

How much do you know about the science and technology in modern China?



➤➤➤ In-class Flipped Teaching : **Step 2**



Cloud Computing

Metaverse

Automated payment

Videophone

Holography

Social Audio

Quantum science and technology

Decentralizing

NFT

云计算

元宇宙

无感支付

可视电话

全息摄影术

语音社交

量子科技

去中心化

非同质化代币





01

**Chinese
Bridges**
中国桥



02

**Chinese
Ports**
中国港



03

**Chinese
Net**
中国网



04

**Chinese
Roads**
中国路



05

**Chinese
Energy**
中国能源





1. 拱形桥 arch bridge
2. 超级工程项目 super engineering projects
3. 基础设施 infrastructure
4. 纵横交错的桥梁网络 a crisscross network of bridges
5. 南水北调工程 South-to-North Water Diversion
6. 西气东输工程 West-to-East natural gas transmission
7. 珠江三角洲 the Pearl River Delta
8. 大数据时代 the era of big data
9. 5G牌照 the 5G license plate



汉译英

1. 一桥飞架南北，天堑变通途。

Many bridges now fly to span the north and south, changing a deep chasm into a thoroughfare.

2. 这一项又一项超级工程，不仅体现了千百年来中国智慧，也谱写了新时代中国的奇迹。

All these super engineering projects _____.

All these super engineering projects, not only **manifest** the Chinese wisdom with a history of thousands of years, but also **show** a miracle of China in the new era.



3. 中国援助项目也正在为我们的全球大家庭建设更美好的未来。

China-aid projects are also helping to build a better future for our families around the world.

4. 尊重自然、顺应自然，这是隐藏在中国人生命里的哲学道理。

Show respect to the nature and be adaptive to the environment is a philosophy hidden in Chinese people's life.

5. 我们不止要掌握这些井喷般增长的数据，更要将它们融合加工，这就是大数据技术的核心意义。

We should not only master these blowout growth data , but also integrate them into processing , which is the significance and value of big data technology .



英译汉

The Suzhou port in the Yangtze River Delta is the world's top port in the connection of rivers, lakes and seas. The three international deep-water ports in the Pearl River Delta work together to contribute to the same aim: “Let the world fall in love with products made in China” .

长江三角洲的苏州港为江海河联运全球之首，珠江三角洲的三个国际深水良港携手努力，都为世界爱上“中国造”贡献了重要的力量。

When the imaginary life is coming true in real life and the new technology is becoming commonplace, the Chinese artificial intelligence technology, which is supported by big data and characterized by 5G speed, is making people's life better and better!

当想象中的生活正一点点在现实世界上演，当新兴科技正变得司空见惯，以大数据为支撑、5G 速度为特色的中国人工智能技术正在将我们的生活变得更加美好！



Chinese Bridges

20



A. Hong Kong-Zhuhai-Macau Bridge

B. the Dankunte bridge

C. Chaotianmen Bridge

D. Beijing-Shanghai high-speed railway bridge

 B is the world's longest high-speed railway bridge.

 C is the world's longest arch bridge.

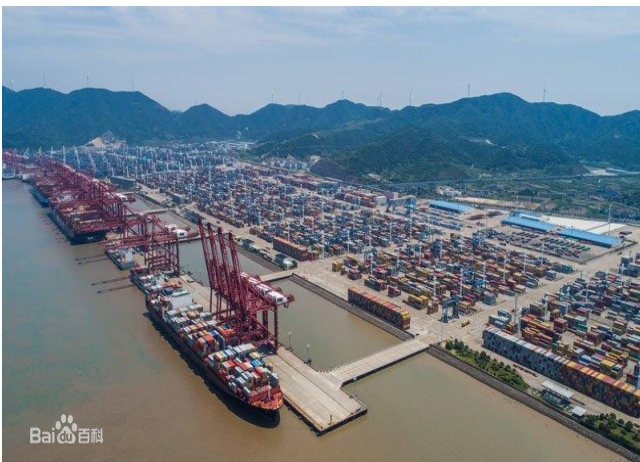
 A is a longest cross-sea bridge in the world.

 D is the world's first six-line railway bridge.



Chinese Ports

21



Zhoushan Port



Suzhou Port



Yangshan Port



Tianjin Port

The B in the Yangtze River Delta is the world's top port in the connection of rivers, lakes and seas.

D is the world's largest artificial deep-water port.

The throughput of A broke the record and has become the world's first port with 900 Million tons throughput in 2016.

C is the world's most highly automated container terminal.

Chinese Roads

The Chinese roads outline a new blueprint for China's urban and rural development.

Jingxin expressway is the world's longest expressway across the the desert. There is also

Yaxi expressway that runs through multiple seismic zones. The average elevation rises 7.5 meters per kilometer and Yaxi expressway is just like a giant dragon in the mountains.

So it is also called the Chinese heaven road. China's highway network ranks first in the world with a total mileage of 131000 kilometers. In 2020, the total length of China's expressway will reach 150000 km.

Chinese Energy

All the 9.6 million square kilometers of land are the places where the Chinese make every effort to explore. Apart from the well-known South-to-North Water Diversion, the West-to-East natural gas transmission, let's have a look at a special farm in Inner Mongolia Autonomous Region: “Solar Farm”.



Can you make a guess what the “Solar Farm” looks like?



What is the core technology applied in the “farm”?

Now, we know the South-to-North Water Diversion, the West-to-East natural gas transmission, the solar farm. Can you introduce the following mega-projects in English?



Beijing Daxing International Airport



FAST



Hong Kong-Zhuhai-Macao Bridge



High Speed Rail

补充: Hangzhou Olympic Sports Center 杭州奥体中心的科技创新

For many stadiums and arenas, there is frequently a tradeoff(折中) between sustainability, performance and iconic design. Hangzhou Sports Center unites all three, featuring a poetic, floral-inspired design that simultaneously reduces carbon and material use. The development uses 67% less steel than Beijing Olympic Stadium, making it both beautiful and sustainable while creating a world-class sporting venue for a booming city.





超级工程我来说

Besides the mega-projects mentioned in our course, what new mega-projects in China do you know? Each group chooses a new project that was constructed recently and makes a video introducing it in 3 minutes.

请介绍下除本课程之外，你了解到近年来中国国内或援外建设的超级工程，英文介绍，可以配图片，100-300字。

陈柳伊同学作业 (Shanghai Tower)

上海中心大厦. Shanghai Tower.

Located in Shanghai, the Shanghai Tower is the tallest building in China. It is mainly used for office, hotel, business, sightseeing and other public facilities. The ^{main} building is 127 floors above ground, which is 632 meters high, and the basement has 5 floors. The total construction area is about 578000 square meters, of which the above-ground area is about 410000 square meters, the underground area is about 168000 square meters, covering an area of 30368 square meters.

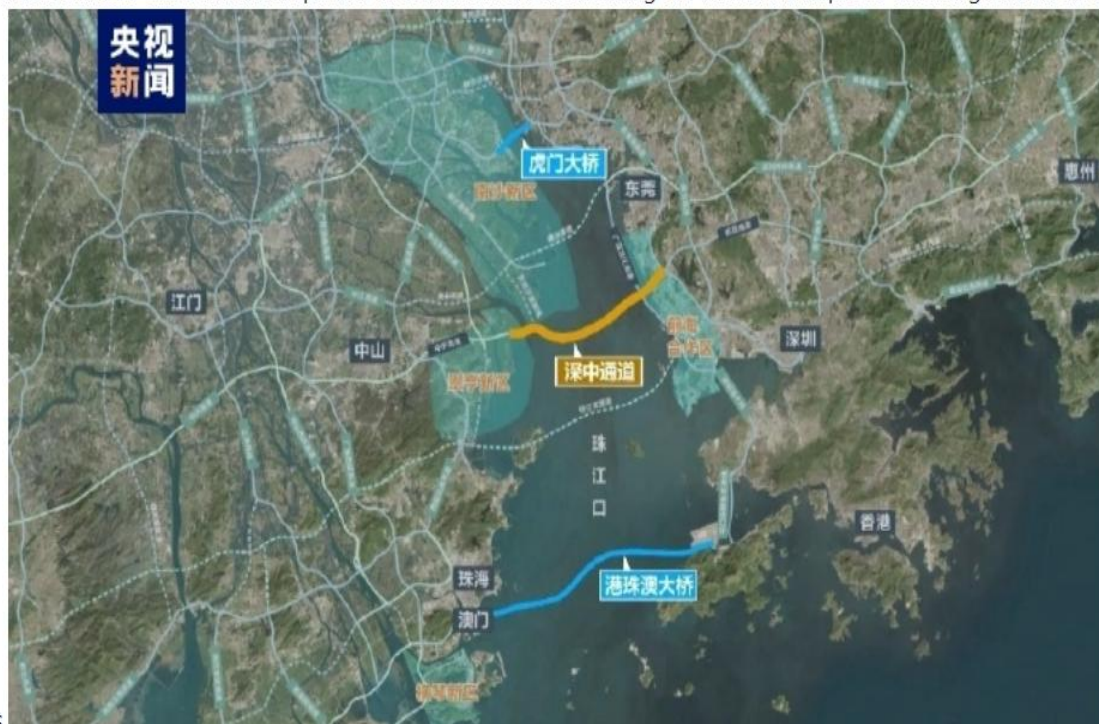
Due to the soft soil in Shanghai, the designers designed 955 foundation piles measuring 86 meters long. For stability, a foundation pit with a diameter of 121 meters and a depth of 33 meters was dug. Most distinctive of all, it has a huge rainwater collection tank ~~on~~ and 15 wind turbines on top. With a 1000-ton damper on the 126th floor, the building is impervious to earthquakes and typhoons.

The Shanghai Tower is in a spiral form as a whole. It looks ~~like~~ like "it is breaking out of the ground" from the ground, like a giant dragon soaring into the sky, implying the take-off of modern China.



石雅婧同学作业 (深圳-中山主线)

After class, I learned that there are still many mega-projects in China in recent years, such as the Jinsha River Bridge, and the China Mobile Intelligent Computing Center (Hohhot). I will detail one of the recent China Bridge projects. The Shenzhen-Zhongshan Channel started construction in December 2016 and the main line was fully connected in November 2023. It is a project that integrates "bridges, islands, tunnels, and underwater interchanges", with the characteristics of "east tunnels and west bridges". The total length is about 24 kilometers, including a 6.8 kilometer long immersed tube tunnel, which is currently the longest and widest underwater immersed tube tunnel in the world, with a design speed of 100 kilometers per hour. Connecting Shenzhen and Zhongshan, commuting between the two places takes only half an hour, improving transportation efficiency. It will promote the connectivity of urban agglomerations, strengthen the economic links between regions, and promote the coordinated development of industries. It will also drive the development of tourism in the areas along the route and improve the living standards of local



褚懿佩同学作业（中欧班列 China-Europe Railways）

The China-Europe Railways is a significant transcontinental railway network that connects China with Europe, passing through Central Asia and Russia. This project aims to improve connectivity and trade between China and Europe, providing a more efficient and cost-effective transportation option compared to traditional shipping routes.



The construction of the China-Europe Railways began in the early 2000s and has been gradually expanded and improved over the years. The railway has significantly shortened the transportation time for goods between China and Europe, making it more attractive for businesses to use this mode of transportation. It has also opened up new opportunities for trade and cooperation between China and Europe, promoting economic growth and development in both regions.

In addition to the transportation of goods, the China-Europe Railways also play an important role in promoting cultural exchanges between China and Europe.

The railway has become a popular mode of transportation for tourists, allowing people to experience the beauty and history of both regions.

Overall, the China-Europe Railways is a remarkable engineering achievement that has significantly improved connectivity and trade between China and Europe. It is a symbol of cooperation and mutual benefit between China and Europe, promoting economic growth and cultural exchange.

吴珂尔同学作业（中欧班列 China-Europe Railways）

The Three Gorges Dam, a monumental engineering feat completed in 2006, spans the Yangtze River in China. As the world's largest hydropower project, this structure stands at approximately 607 feet (185 meters) tall and extends over 1.4 miles (2.3 kilometers) in length.

Its primary objectives encompass flood control, electricity generation, and improving navigability along the Yangtze River. With a reservoir capacity exceeding 39 cubic miles (160 cubic kilometers), the dam significantly mitigates the historically devastating floods that plagued the region.

This impressive engineering endeavor generates substantial hydroelectric power, contributing significantly to China's renewable energy production. However, the project hasn't been without controversy due to concerns over environmental impact, displacement of communities, and geological implications.

Despite these challenges, the Three Gorges Dam symbolizes China's commitment to sustainable energy and flood management. Its monumental size and multifaceted functions stand as a testament to China's engineering prowess while grappling with the complex balance between economic development and environmental preservation.



何婧同学作业（Karachi Nuclear Plant In Pakistan）

Karachi Nuclear Power Plant in Pakistan

The Karachi Nuclear Power Plant, located in Karachi, Pakistan, stands as a testament to the strong bilateral ties between Pakistan and China. Spearheaded by the China National Nuclear Corporation (CNNC), this colossal undertaking represents a significant stride in the field of nuclear energy.

Key Features:

Collaborative Endeavor: Established through a collaboration between China and Pakistan, the Karachi Nuclear Power Plant symbolizes the synergy achieved through international cooperation in advancing nuclear technology.

Cutting-edge Technology: Employing state-of-the-art nuclear technology, the power plant adheres to stringent safety standards, ensuring a reliable and secure source of energy for the region.

Capacity and Impact: As one of the largest nuclear power plants in Pakistan, it significantly contributes to the nation's energy infrastructure, addressing the growing demand for electricity and fostering economic development.

Strategic Importance: The strategic location of the Karachi Nuclear Power Plant enhances energy security, playing a vital role in meeting the power needs of Karachi, the economic hub of Pakistan.

This monumental project not only underscores the technical prowess of the China National Nuclear Corporation but also highlights the shared commitment of both nations to sustainable development through the responsible use of nuclear energy. The Karachi Nuclear Power Plant stands as a beacon of progress, providing a blueprint for future collaborations in the global pursuit of clean and efficient energy solutions.

巴基斯坦卡拉奇核电站

位于巴基斯坦卡拉奇的核电站是中巴两国坚实双边关系的象征。由中国国家核能公司（CNNC）领导，这一巨大的项目代表了核能领域的重要进展。

主要特点：

协同合作：卡拉奇核电站是中巴两国合作的成果，象征着通过国际合作推进核技术所取得的协同效应。

尖端技术：采用先进的核技术，该核电站符合严格的安全标准，确保为该地区提供可靠且安全的能源来源。

容量和影响：作为巴基斯坦最大的核电站之一，它在国家能源基础设施中发挥着重要作用，满足了电力需求的增长，促进了经济发展。

战略重要性：卡拉奇核电站的战略位置增强了能源安全性，对满足卡拉奇作为巴基斯坦经济中心的电力需求起到了至关重要的作用。

这一巨大的项目不仅彰显了中国国家核能公司的技术实力，也突显了两国共同致力于通过负责任的核能利用实现可持续发展。卡拉奇核电站是进步的象征，为未来全球寻求清洁高效能源解决方案的合作提供了范本。

何婧同学作业 (Karachi Nuclear Plant In Pakistan)



韩湘楠同学作业 (The Peljesac Bridge)

The Pelješac Bridge is a significant infrastructure project in Croatia. The bridge aims to connect the Pelješac Peninsula to the mainland, bypassing the need to cross a short strip of Bosnia and Herzegovina's territory. This project is important for improving regional connectivity and reducing travel time. The bridge is designed to be approximately 2.4 kilometers long, making it one of the longest bridges in the country.



1. 自学国务院新闻办公室的《中国的对外援助》（英文版）

http://www.scio.gov.cn/ztk/xwfb/31/8/document/899557/899557_7.htm



汉译英

1. 在过去，“中国制造”往往与低价格、低质量联系在一起，这使人们关注的是数量而不是质量。

In the past, products "made in China" was often associated with things of low price and poor quality, which draw people's attention to quantity rather than quality.

2. 同时，中国也已经成为全球人工智能专利最多的国家，拥有世界第二的人工智能技术人才数量。

At the same time, China has also become the country with the largest number of artificial intelligence patents in the world, with the second largest number of artificial intelligence technical personnel in the world.

3. 手机，时时刻刻陪伴我们生活左右，中国以每年接近 20 亿部智能手机的产量成为当之无愧的手机制造大国。

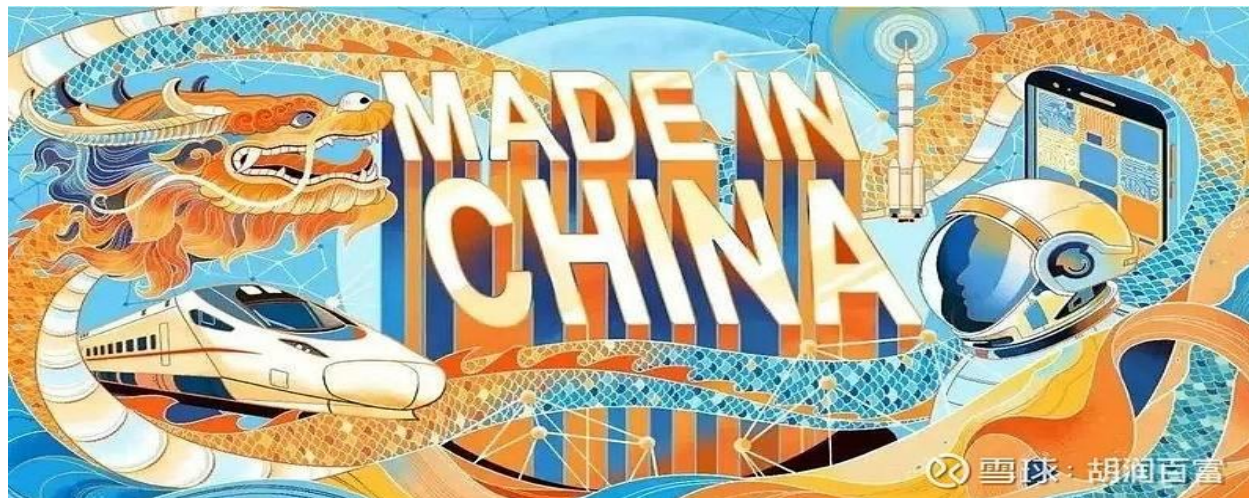
Mobile phones accompany us around our lives all the time. China has become a well-deserved smart phone manufacturing country with an annual output of nearly 2 billion smartphones.

4. 在中国古代，嫦娥奔月的传说反映了人们对遥远宇宙的向往。50年来，中国人在航天领域进行了不懈的探索，取得了重要成就。

In ancient China, the legend of Chang'e running to the moon reflects people's yearning for the distant universe. In the past 50 years, the Chinese have made unremitting explorations in the field of space and made important accomplishments.

Questions:

1. As the generation after 00s, what's your first reaction to products "Made in China"?
2. What's the difference between "Made in China" and "Intellectually Manufactured in China"? Please cite some examples.



Made in China
has swept

Could you
overseas demand
its popularity

ring brand

popularity
the reason for



For quite a long time, even today, the image of Chinese-made products was often perceived as cheap and low-end.



However, have you ever wondered:

What negative consequences would the export industry only producing low price and poor quality commodity bring for China's economic development?

1. By relying on low-price products to open overseas markets, China's manufacturing industry is only at the end of the global industrial chain.

2. Becoming the biggest contract manufacturer in the world(世界最大代加工厂) China will consume a lot of domestic natural resources and serious environmental pollutions would follow up.

3. Cheap exports expose Chinese enterprises to a range of foreign anti-dumping treatment(反倾销), which makes them suffer huge losses.

4. From the micro perspective, labor costs are low and hard work is not well paid. Here are 2 examples.

以芭比娃娃与iPad这两个极具代表性的“代工产品”为例。

世界上大部分芭比娃娃都由中国制造，若以10美元一个售价来计算，拥有该品牌的美国获利8美元，而中国企业只能拿到35美分的加工费。

标有“Made in China”的iPad在美国最低售价约499美元一台，但每卖出一台iPad，富士康和深圳工厂仅得9美元。由代工芭比娃娃到代工iPad，“中国制造”的代工技术显著升级，但中国从中获取的毛收益比重却在下降。



In the competition of the international division of labor system, the former "world factory" America, Britain and Japan occupy the high-end of the international division of labor chain.

Therefore,

It is imperative for China to think ahead and move from "Made in China" to "Intellectually Manufactured in China" and change from "the biggest contract manufacturer" to "the world creation base".



CCTV 13

新闻

CCTV.com

数字十年

新闻直播间
LIVE NEWS

大国重器挺起中国制造脊梁

10月22日 16:44

明天夜间至25日，海南岛等地局地暴雨，南海海域风力较大。

CCTV 新闻 俄



盘点那些让我们热血沸腾的大国重器

你能说出几个？

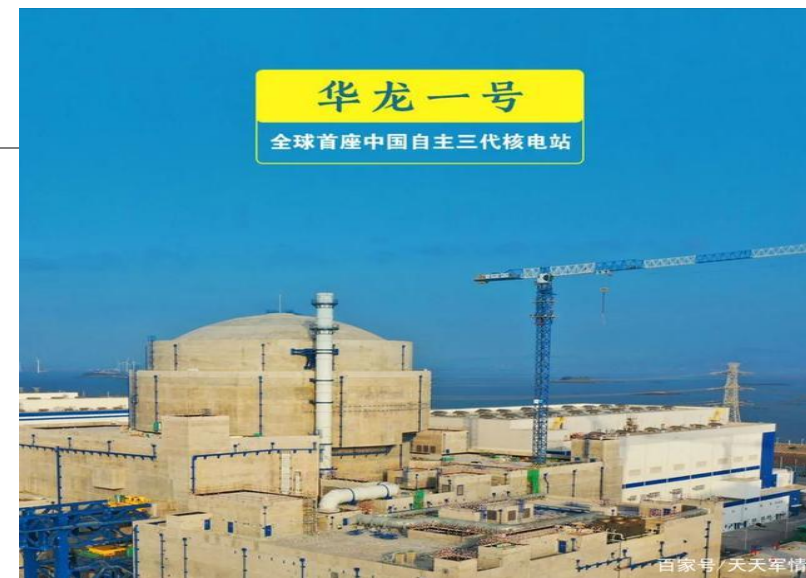




天问一号火星探测器



全海深载人潜水器“奋斗者”号



华龙一号



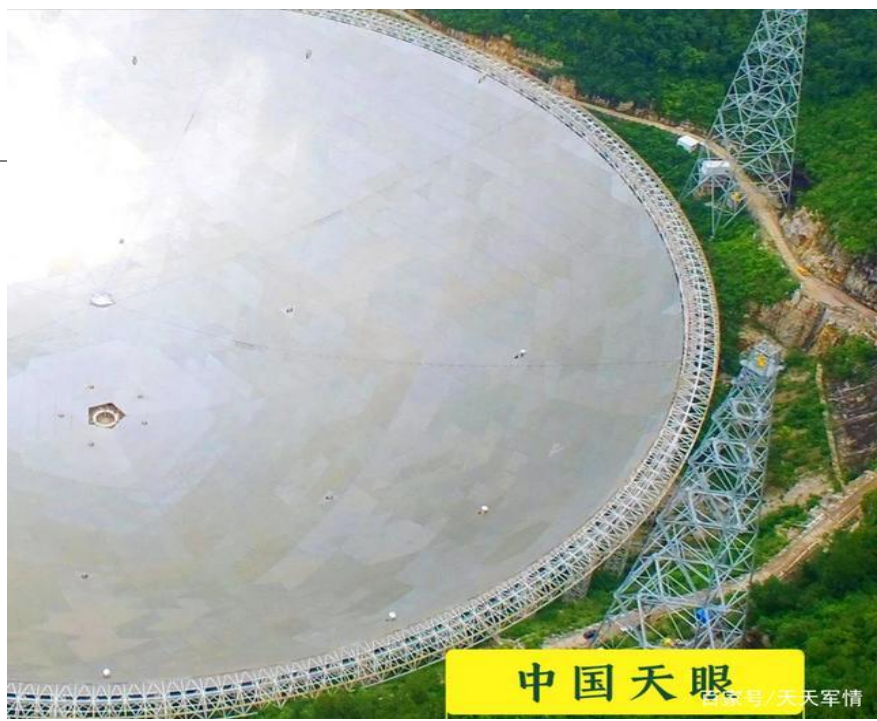
北斗卫星导航系统



AG600鲲龙大型水陆两栖飞机



中国人民解放军海军山东舰



China to Catch up with the World in Artificial Intelligence

foreseeable gap core designated strategic unique surpass

In recent years, China's _____ strength in Big Data is playing an increasingly important role in the 4th industrial revolution, the _____ of which lies in artificial intelligence.

The government has _____ artificial intelligence as a _____ area. With the strong support of the government, the AI industry is attaining rapid development in China. It is already narrowing the _____ with the US, and can be expected to _____ the US in the _____ future.

The performance of China's AI companies in algorithms, computing power and data analysis has significantly improved in recent years. 9 Chinese companies – including IFLYTECK, Tencent and BAT – have been listed in the 50 Smartest Companies of 2017 by the MIT Technology Review. 《麻省理工科技评论》

China to Catch up with the World in Artificial Intelligence

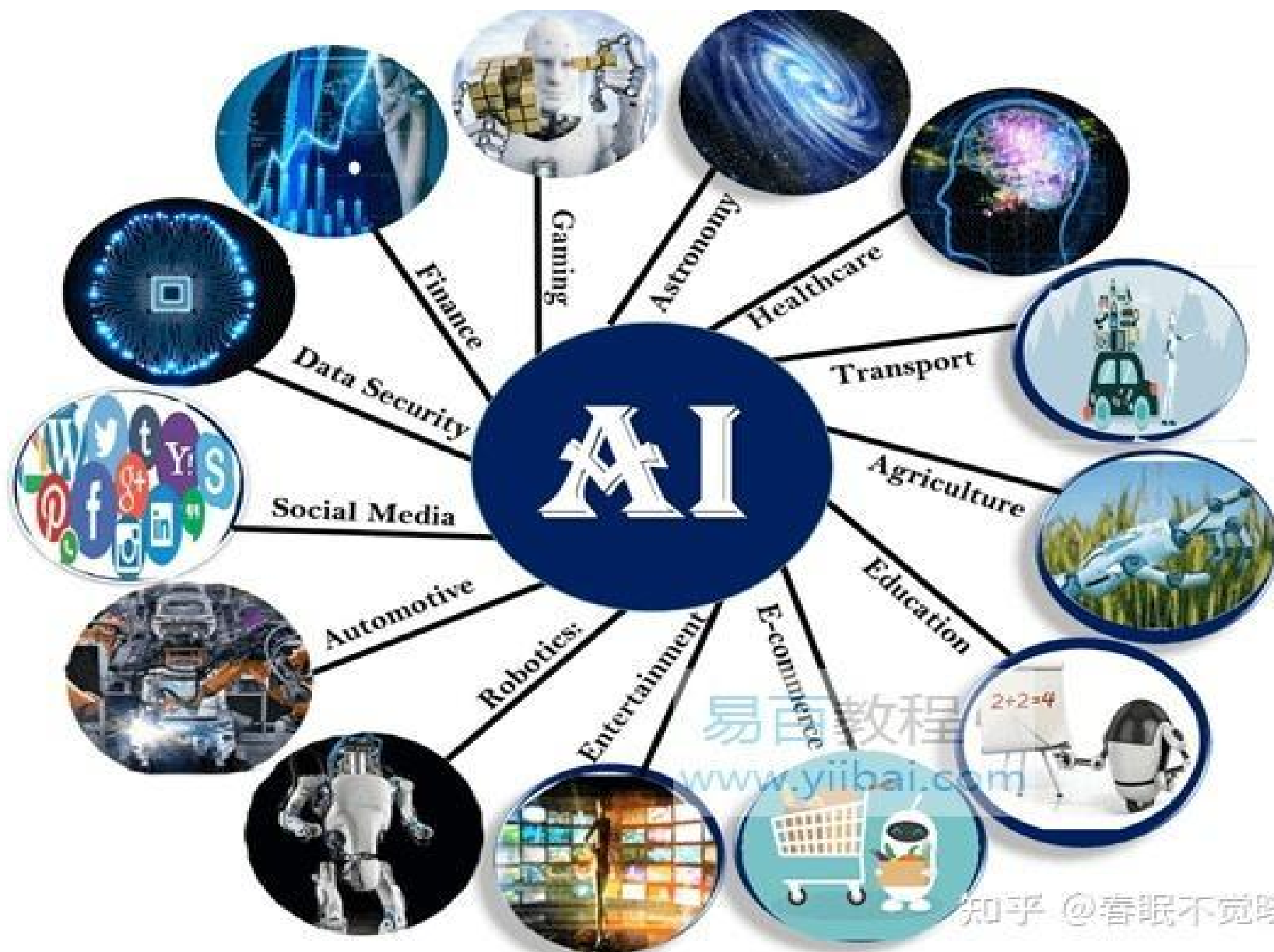
In recent years, China's **unique** strength in Big Data is playing an increasingly important role in the 4th industrial revolution, the **core** of which lies in artificial intelligence.

The government has **designated** artificial intelligence as a **strategic** area. With the strong support of the government, the AI industry is attaining rapid development in China. It is already narrowing the **gap** with the US, and can be expected to **surpass** the US in the **foreseeable** future.

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>>> The application fields of AI in China Can you name some?

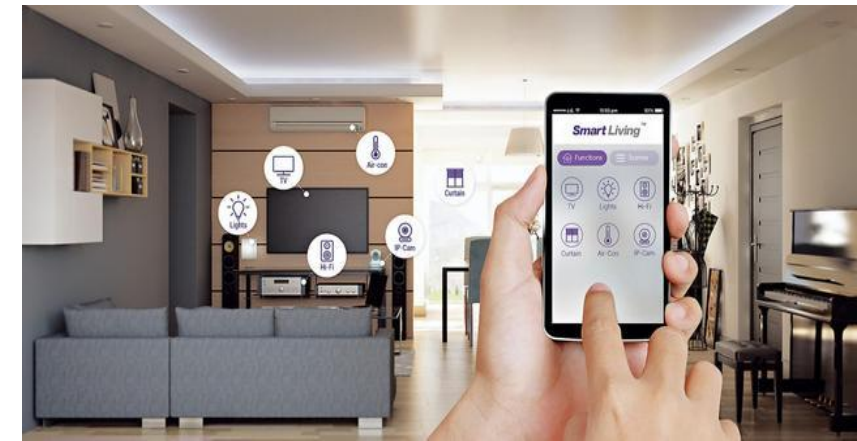
51



AI and smart home

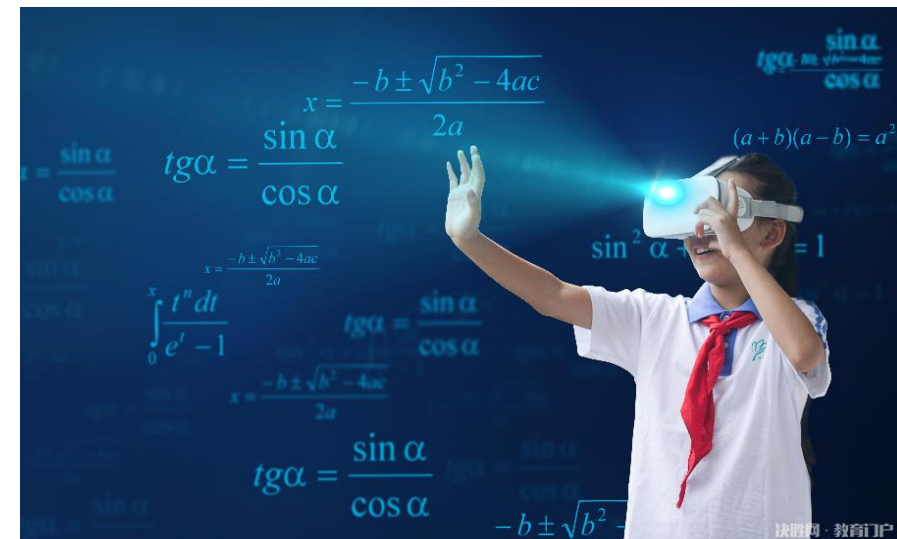
Smart speakers, smart TVs, smart door locks, smart scene panels now enter many households. Users can control smart home products by speaking, such as to switch curtains, turn on and off lighting system, adjust volume, switch TV programs and so on.

With the help of machine learning and deep learning technology, intelligent TV, intelligent speaker and so on can recommend the content that the user may be interested in according to the historical data subscribed or watched by the user. The combination of imagination and technology has made us the practitioners of intelligent life.



AI and education

Now, in China, the new wisdom classroom is slowly coming to us, combining multimedia and web-based technology. With the help of Internet of things technology, cloud computing technology and intelligent devices, teachers can obtain students' pre-class learning data in a timely manner, and develop personalized teaching programs. In class, teachers and students can also interact based on intelligent teaching tools. At the same time, there are various types of online classroom resources waiting for you to explore.



Now, it's your turn to design some AI products in the designated field. Each group is given the specified topic and is required to work out at least one AI product in that field.

You need to:

1. Give your AI product a wonderful name
2. Illustrate its main functions
3. Figure out an impressive advertising slogan

AI and transportation

AI and business marketing

AI and e-commerce

AI and entertainment

AI and cooking

AI and healthcare

AI and security

AI and social media

Watch the following video : **Amazing China**

Discuss with your partner and list the possible reasons for the brilliant achievements on Science and Technology China has made in the past decades. You can surf the internet and find out more reasons if they are not mentioned in this video.



搜狐视频
tv.sohu.com

Amazing China:
A 5-year Miracle

5 YEARS

1. Government support

science and technology are viewed as the cornerstone of a powerful nation, a foundation on which creative talents can build.

Three-step goals: To **develop China** into an innovation-oriented country by 2020, to **place China** among top innovation-oriented countries by 2030, and to **turn China into** a technological innovation **powerhouse** by 2025. (翻译成中文)

三步走战略目标：第一步，到2020年进入创新型国家行列；第二步，到2030年跻身创新型国家前列；第三步，到2050年建成世界科技创新强国。



2. S&T investment

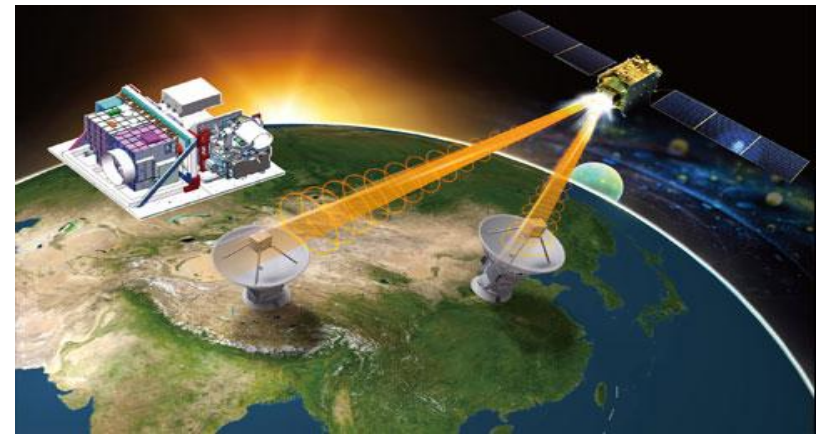
Over the past decade, China's S&T investment has substantially increased, and the gross domestic R&D expenditure has grown from 1.03 trillion yuan to 2.79 trillion yuan, **ranking second in the world**. The R&D intensity has increased from 1.91% to 2.44%, close to the average level of OECD countries. The basic research expenditure is 3.4 times that of ten years ago, and its share in total R&D expenditure is estimated at 6.09%, hitting a record high.



3. S&T workforce

We have underscored the role of **high-caliber professionals** (高层次专业人才) in driving development, and cultivated a large-scale, well-structured and vibrant S&T workforce. The S&T workforce continues to grow, and the total number of R&D personnel in 2021 stood at 5.62 million, 1.7 times that of 2012, remaining the largest in the world. The number of R&D personnel per 10,000 employees is estimated to increase from 42.6 in 2012 to 75.3.

科技队伍持续壮大，2021年研发人员总数达到562万人，是2012年的1.7倍，继续居世界首位。每万名员工拥有的研发人员数量预计将从2012年的42.6人增加到75.3人。





作品创造： Scientific innovations in 120 seconds

60

Design a 120-second short video to introduce one of China's scientific innovations to an international audience. 利用AI制作一则“中国科技创新”主题的2分钟英文新闻播报。

第一部分： Your design should include the following four aspects. 1) **Target Audience** (e.g., international students, digital nomads, cultural tourists, global social media users); 2) **Content Elements** (e.g., Chang'e, Shenzhou, Tiangong, Mozi, etc.); 3) **Creative Ideas** (e.g., digital storytelling techniques, AI-generated visuals, animation, voiceover narration, interactive elements); 4) **Video Structure** (120 seconds). Provide a brief outline of the video, showing the content plan across the full 120 seconds (e.g., 0–10s, 10–40s, 40–60s). 以上是具体设计与脚本撰写。



撰写“中国科技创新”主题的2分钟英文视频的脚本

- 北斗导航系统
- 高铁出海（如雅万高铁）
- “天问一号” 火星探测
- 量子通信/超级计算
- 绿色科技（如光伏、风能、碳中和）
- 华为芯片突破与5G通信技术

✓ 一、结构框架（适合新闻播报视频/口播任务）

部分	内容	建议长度
开头 (Lead)	引出新闻话题，简洁有力，点明中国科技亮点	1-2句 (30词)
主体 (Body)	介绍具体科技成果：背景、成就、国际意义	3段 (约180词)
结尾 (Closing)	总结意义，展望未来或强调中国创新精神	1段 (40-50词)

（请见钉钉群参考回答框架，请继续补充细节）